TD - Computational Fluid Dynamics : Compressible flows

Abderahmane MAROUF

TD - Computational Fluid Dynamics : Compressible flows

	Conservative form	***Non-conservative form***
PDE:	$\frac{\partial u}{\partial t} + \frac{\partial}{\partial x}(f(u)) = 0$	$\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = 0 \quad , \quad a = \partial f / \partial u$
System of PDE:	$\frac{\partial U}{\partial t} + \frac{\partial}{\partial x}(F(U)) = 0$	$\frac{\partial U}{\partial t} + A\frac{\partial U}{\partial x} = 0 \quad , \quad A = \partial F / \partial U$

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$$\frac{\partial U}{\partial t} + \frac{\partial}{\partial x}(F(U)) = 0$$

with $U = \begin{bmatrix} \rho \\ \rho u \\ \rho E \end{bmatrix}$ and $F(U) = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ (\rho E + p)u \end{bmatrix}$

ρ : density

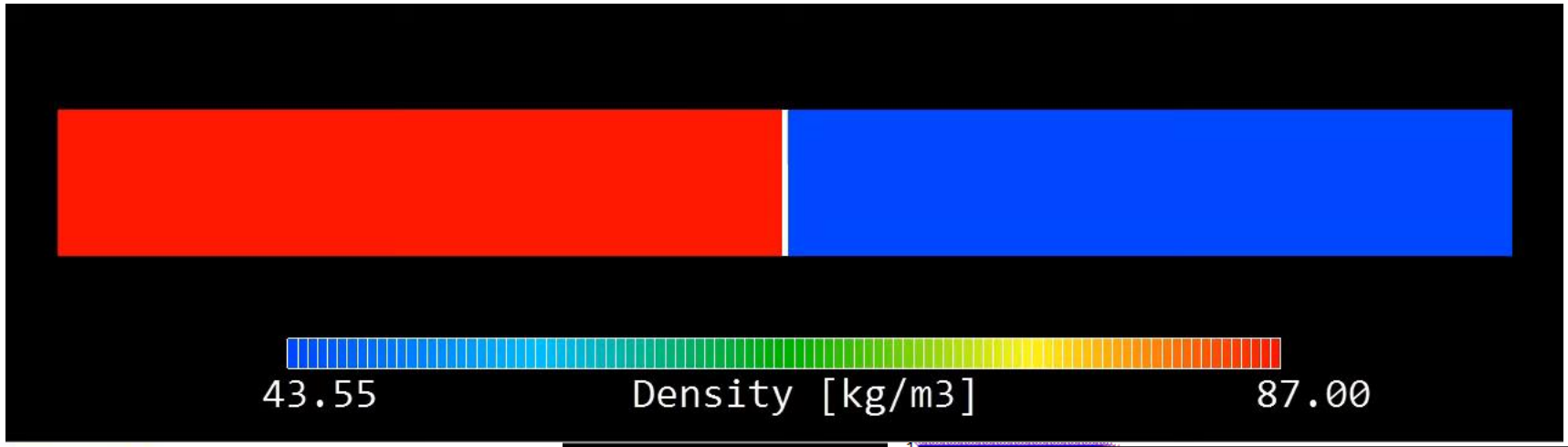
u : velocity

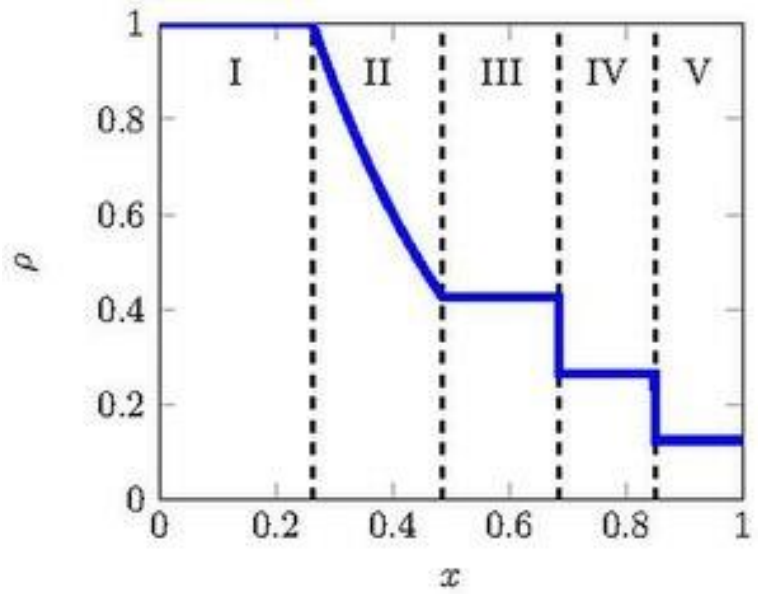
p : pressure

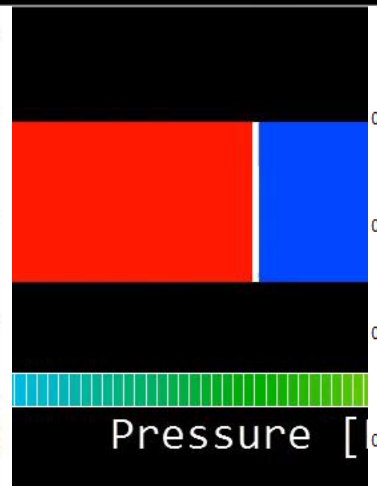
ρE : total energy per unit volume

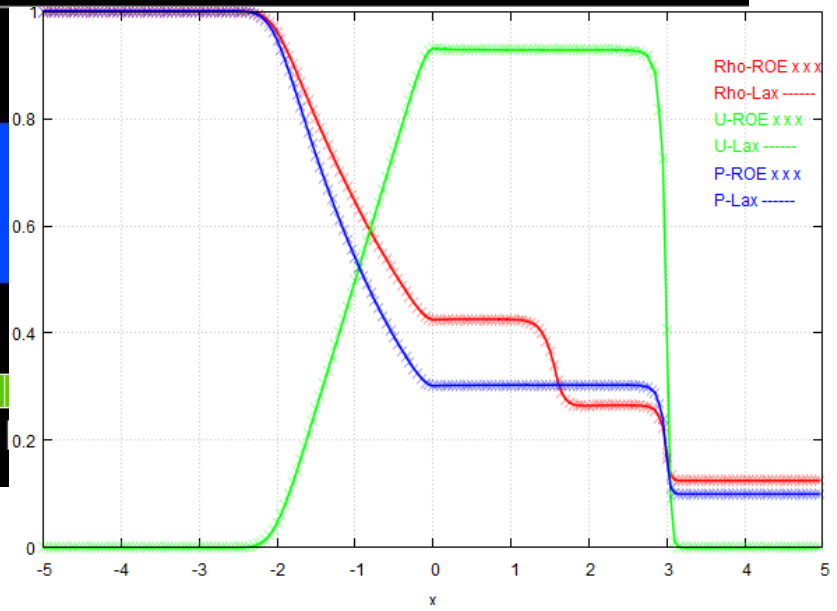
ideal gas: $p = (\gamma - 1)\rho \left[E - \frac{1}{2}u^2 \right]$

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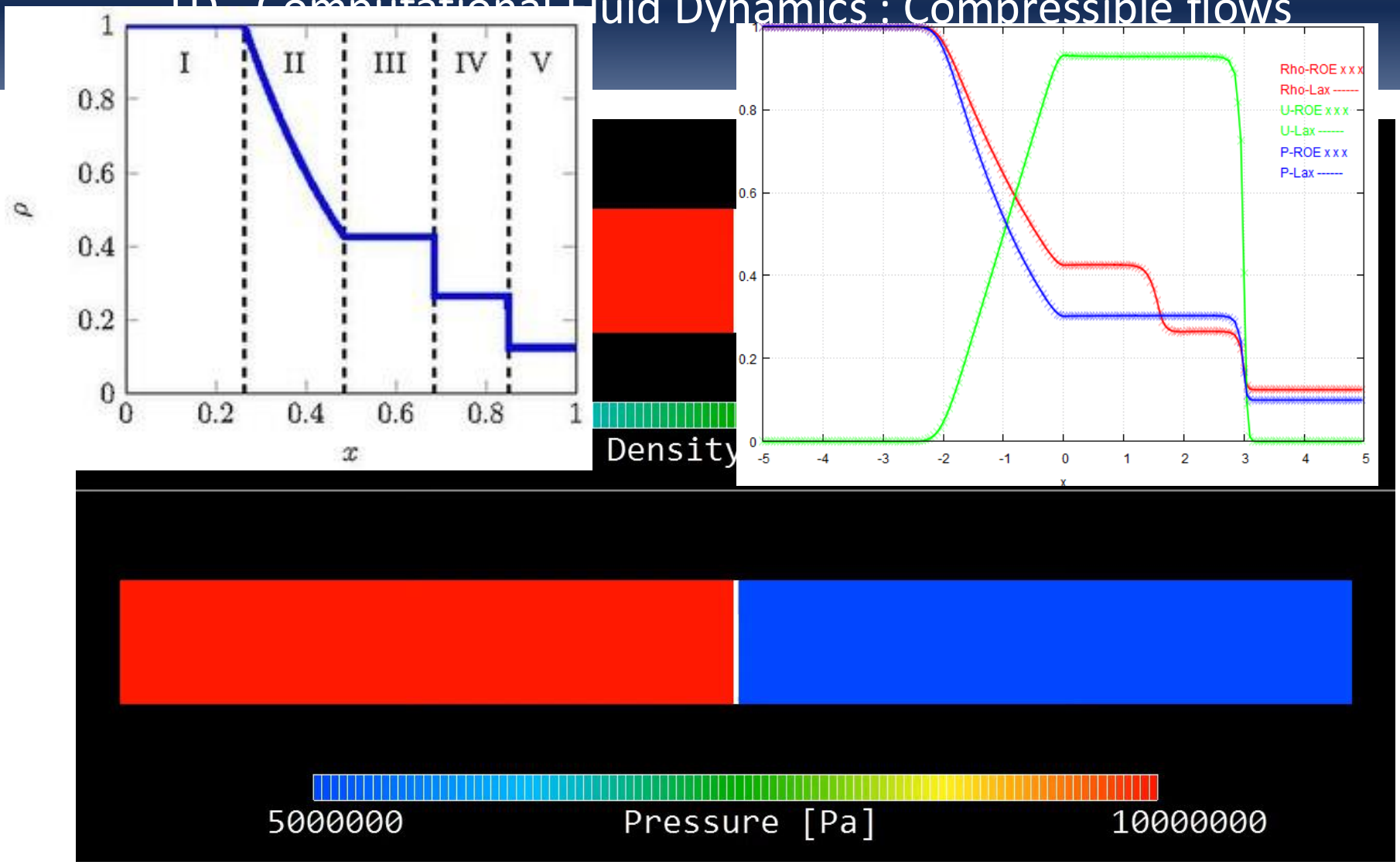


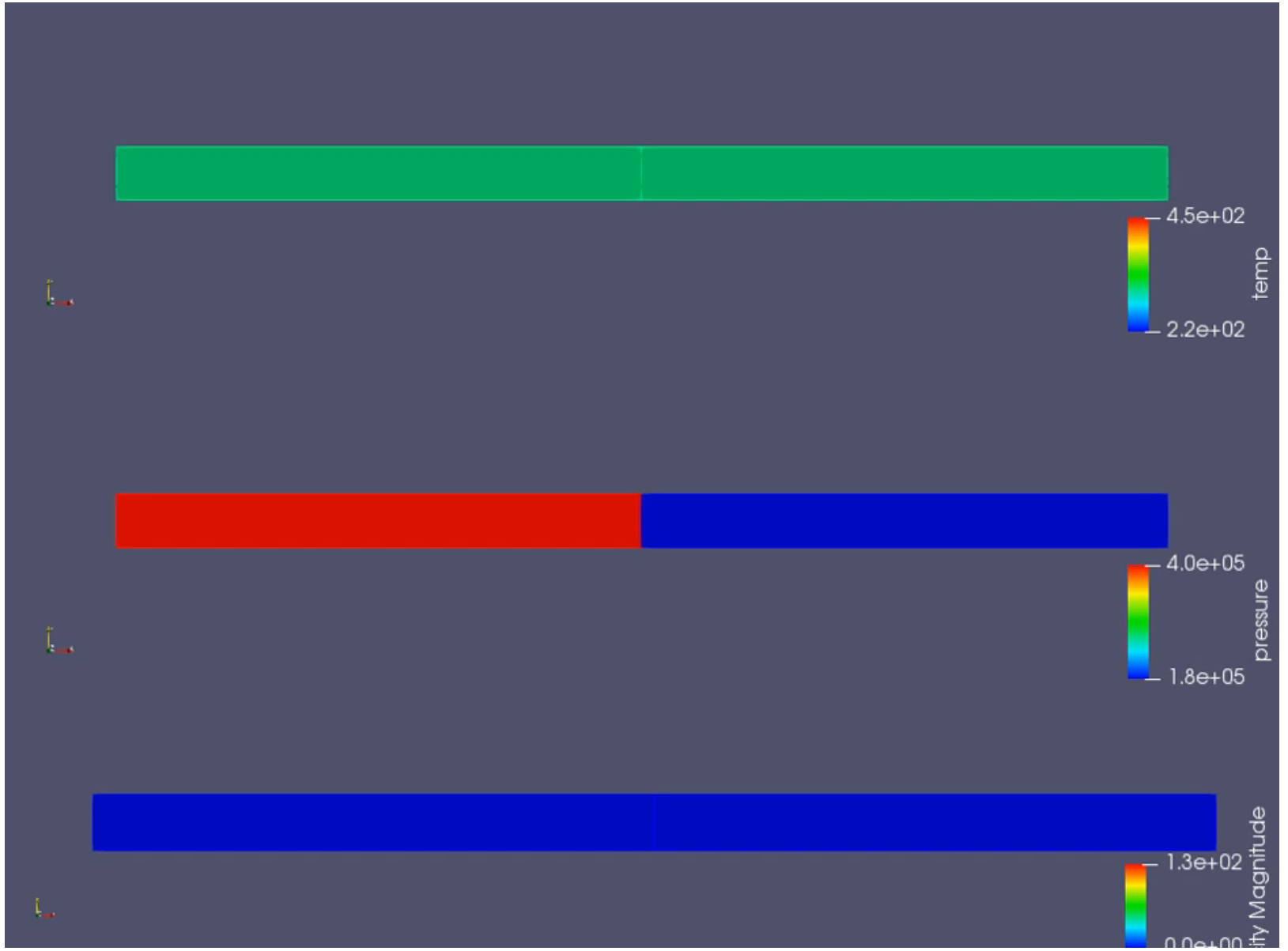




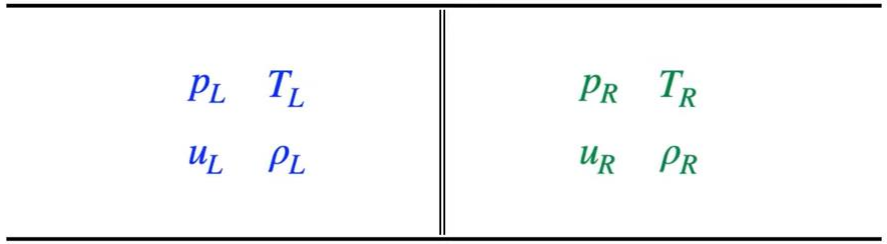


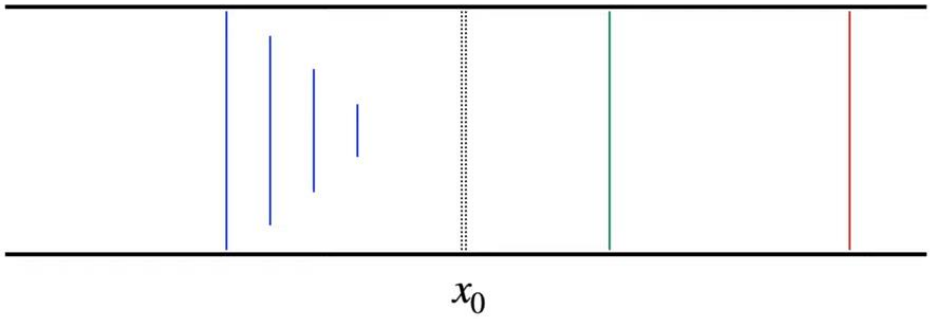
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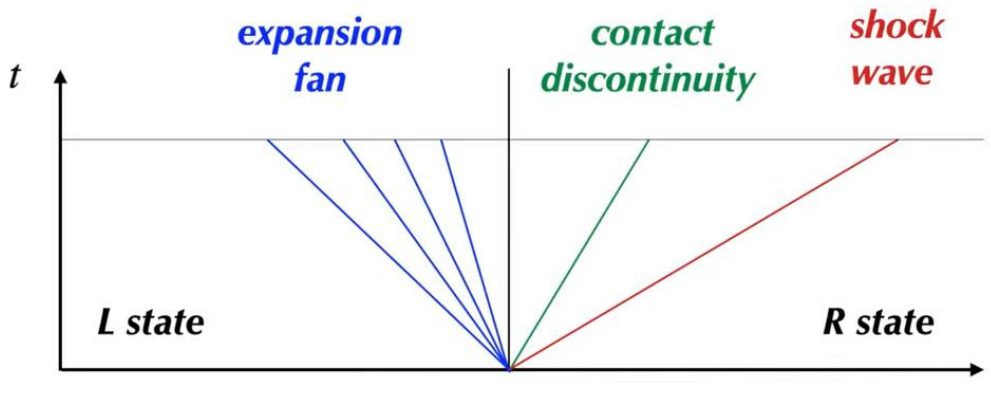


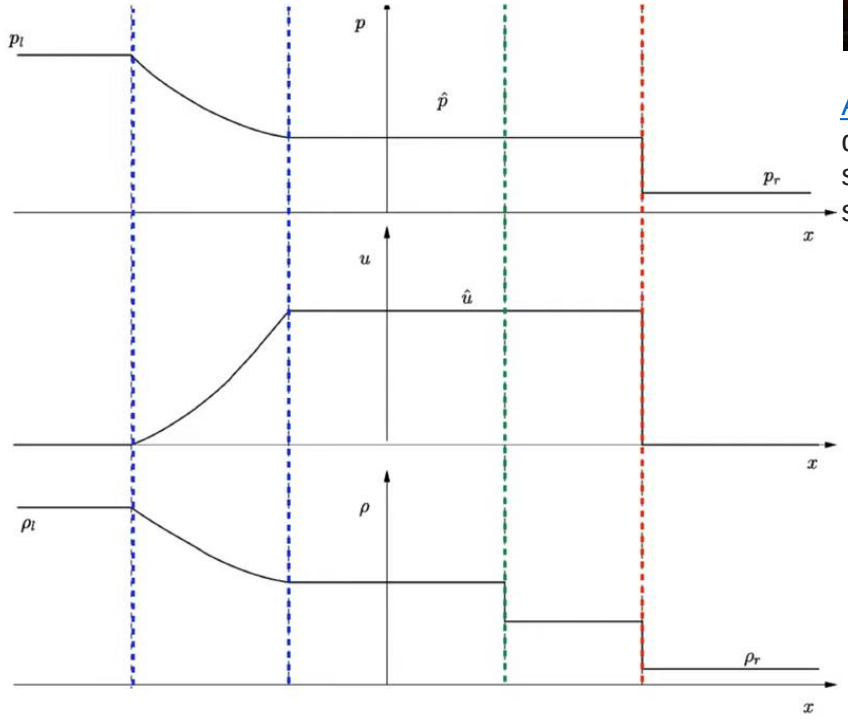

4.5e+02
2.2e+02
temp
4.0e+05
1.8e+05
pressure
1.3e+02
0.0e+00
ity Magnitude

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Aluminium foil used as a diaphragm between shock tube pipe segments

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Initial condition:

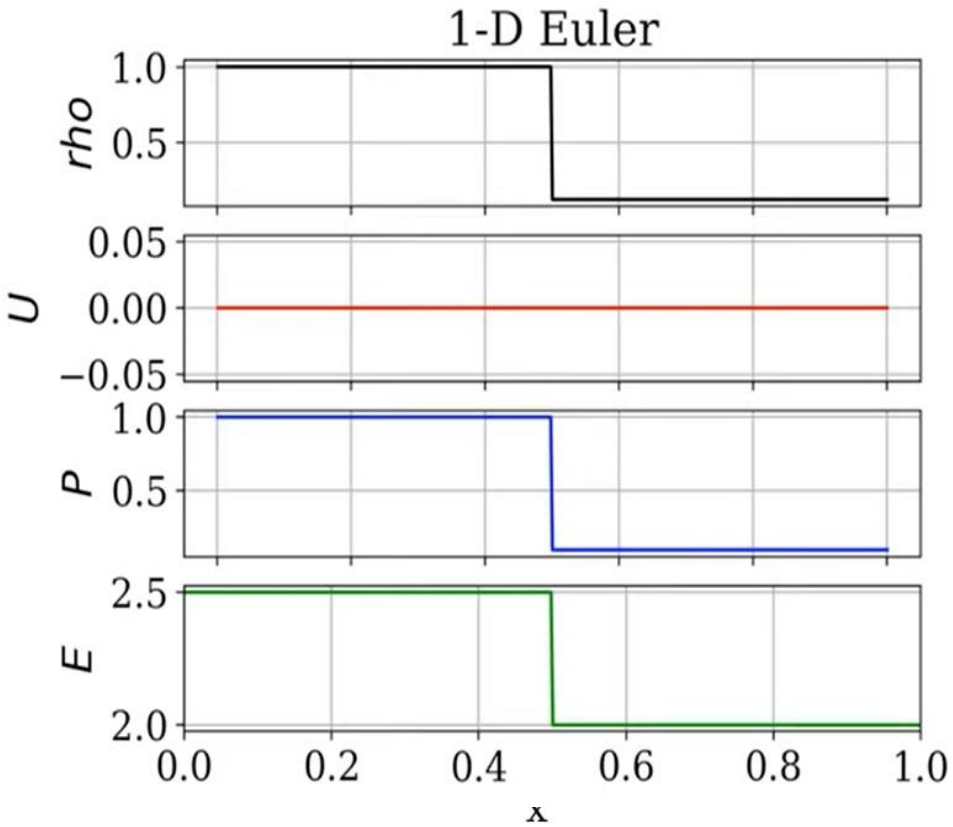
$p_L = 1$	$p_R = 0.1$
$\rho_L = 1$	$\rho_R = 0.125$
$u_L = 0$	$u_R = 0$

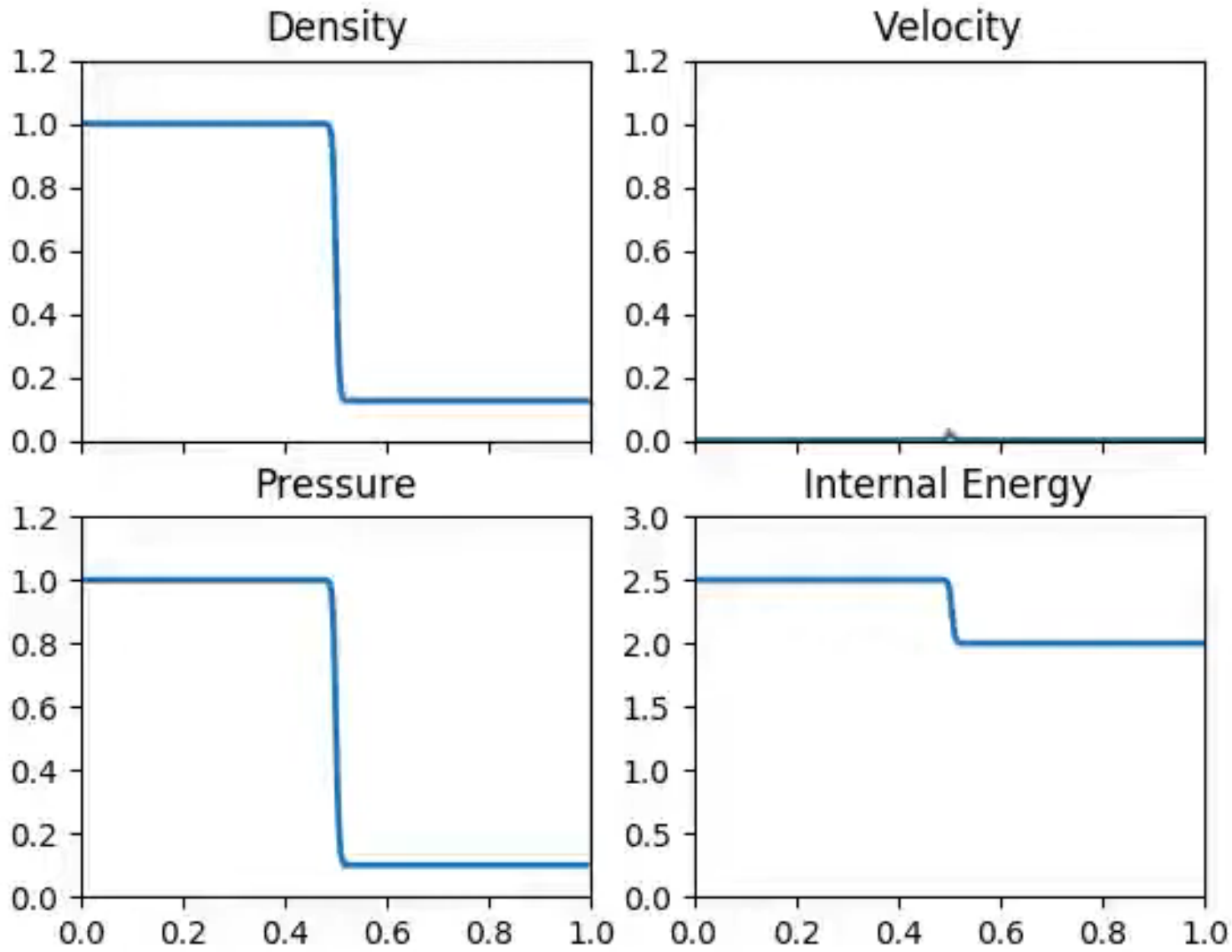
Step 1:
(predictor step)

$$\tilde{U}_{j+1/2} = \frac{U_j^n + U_{j+1}^n}{2} - \frac{\Delta t}{2\Delta x}\left[F(U_{j+1}^n) - F(U_j^n)\right]$$

Step 2:
(corrector step)

$$U_j^{n+1} = U_j^n - \frac{\Delta t}{\Delta x}\left[F(\tilde{U}_{j+1/2}^n) - F(\tilde{U}_{j-1/2}^n)\right]$$

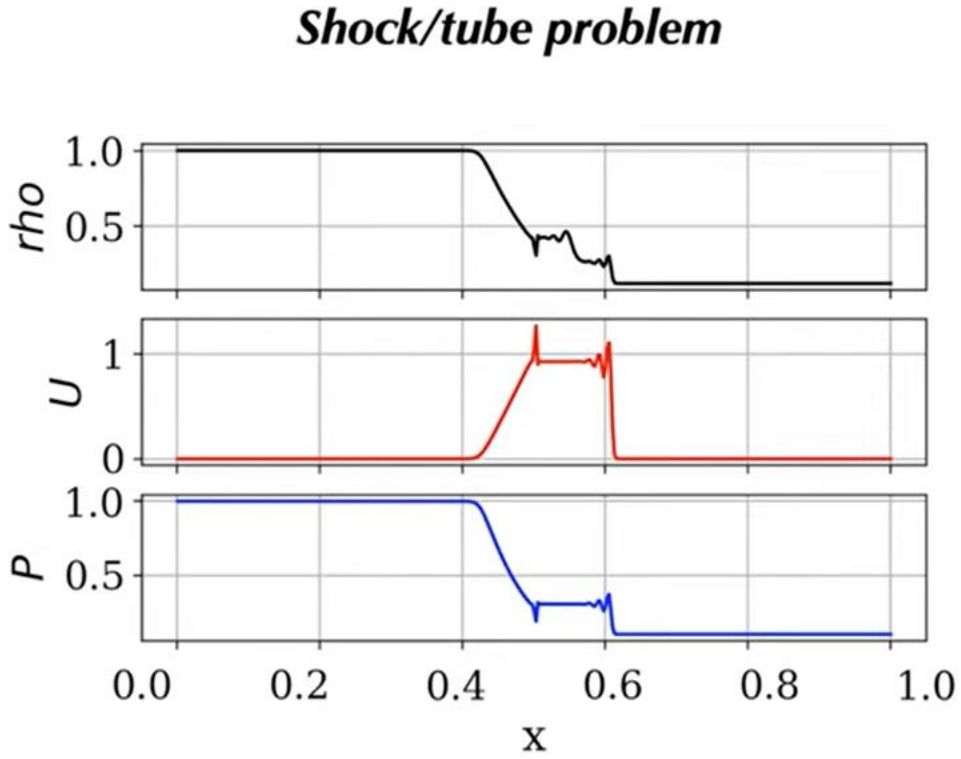



Density
Velocity
Pressure
Internal Energy
1.2
1.0
0.8
0.6
0.4
0.2
0.0
3.0
2.5
2.0
1.5
0.5
0.0
0.2
0.4
0.6
0.8
1.0

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Lax-Wendroff scheme:

(previous video)



Upwind schemes: Godunov's and Roe's schemes

PDE:

$$\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = 0$$

Upwind scheme:

$$u_j^{n+1} = u_j^n - a\frac{\Delta t}{2\Delta x}(u_{j+1}^n - u_{j-1}^n) + |a|\frac{\Delta t}{2\Delta x}(u_{j+1}^n - 2u_j^n + u_{j-1}^n)$$

System of PDE:

$$\frac{\partial U}{\partial t} + A\frac{\partial U}{\partial x} = 0$$

Upwind scheme:

$$U_j^{n+1} = U_j^n - A\frac{\Delta t}{2\Delta x}(U_{j+1}^n - U_{j-1}^n) + |A|\frac{\Delta t}{2\Delta x}(U_{j+1}^n - 2U_j^n + U_{j-1}^n)$$

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System of PDE:

$$\frac{\partial U}{\partial t} + A\frac{\partial U}{\partial x} = 0$$

Upwind scheme:

$$U_j^{n+1} = U_j^n - A\frac{\Delta t}{2\Delta x}(U_{j+1}^n - U_{j-1}^n) + \left|A\right|\frac{\Delta t}{2\Delta x}(U_{j+1}^n - 2U_j^n + U_{j-1}^n)$$

System of PDE:

$$\frac{\partial U}{\partial t} + \frac{\partial}{\partial x}(F(U)) = 0$$

(conservative formulation)

Upwind scheme:

$$U_j^{n+1} = U_j^n - \frac{\Delta t}{2\Delta x}(F_{j+1}^n - F_{j-1}^n) + \frac{\Delta t}{2\Delta x}\left[\left|A_{j+\frac{1}{2}}\right|(U_{j+1}^n - U_j^n) - \left|A_{j-\frac{1}{2}}\right|(U_j^n - U_{j-1}^n)\right]$$

$A = \partial F/\partial U$ ***is the Jacobian matrix***

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$$\frac{\partial U}{\partial t} + \frac{\partial}{\partial x}(F(U)) = 0$$

with $U = \begin{bmatrix} \rho \\ \rho u \\ \rho E \end{bmatrix}$ and $F(U) = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ (\rho E + p)u \end{bmatrix}$

ρ : density

u : velocity

p : pressure

ρE : total energy per unit volume

For a calorically ideal gas: $p = (\gamma - 1)\rho \left[E - \frac{1}{2}u^2 \right]$

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$$\frac{\partial U}{\partial t} + \frac{\partial}{\partial x}(F(U)) = 0 \qquad \text{with} \qquad U = \begin{bmatrix} \rho \\ \rho u \\ \rho E \end{bmatrix} \quad \text{and} \quad F(U) = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ (\rho E + p)u \end{bmatrix}$$

Let's compute the Jacombian matrix $A(U) := \partial F / \partial U$

$$F(U) = \begin{bmatrix} U_2 \\ \frac{1}{2}(3-\gamma)\frac{U_2^2}{U_1} + (\gamma-1)U_3 \\ \gamma\frac{U_2 U_3}{U_1} - \frac{1}{2}(\gamma-1)\frac{U_2^2}{U_1^2} \end{bmatrix} \qquad A(U) = \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{2}(\gamma-3)\left(\frac{U_2}{U_1}\right)^2 & (3-\gamma)\frac{U_2}{U_1} & \gamma-1 \\ -\gamma\frac{U_2 U_3}{U_1^2} + (\gamma-1)\left(\frac{U_2}{U_1}\right)^3 & \gamma\frac{U_3}{U_1} - \frac{3}{2}(\gamma-1)\left(\frac{U_2}{U_1}\right)^2 & \gamma\frac{U_2}{U_1} \end{bmatrix}$$

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$$\frac{\partial U}{\partial t} + \frac{\partial}{\partial x}(F(U)) = 0 \qquad \text{with} \qquad U = \begin{bmatrix} \rho \\ \rho u \\ \rho E \end{bmatrix} \qquad \text{and} \qquad F(U) = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ (\rho E + p)u \end{bmatrix}$$

Let's compute the Jacombian matrix $A(U) := \partial F / \partial U$

$$A(U) = \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{2}(\gamma - 3)\left(\frac{U_2}{U_1}\right)^2 & (3-\gamma)\frac{U_2}{U_1} & \gamma - 1 \\ -\gamma\frac{U_2 U_3}{U_1^2} + (\gamma - 1)\left(\frac{U_2}{U_1}\right)^3 & \gamma\frac{U_3}{U_1} - \frac{3}{2}(\gamma - 1)\left(\frac{U_2}{U_1}\right)^2 & \gamma\frac{U_2}{U_1} \end{bmatrix}$$

$$H = E + \frac{p}{\rho} = \frac{a^2}{\gamma - 1} + \frac{1}{2}u^2$$

$$a = \sqrt{\frac{\gamma p}{\rho}}$$

$$\frac{\partial U}{\partial t} + A\frac{\partial U}{\partial x} = 0 \qquad \text{with} \quad U = \begin{bmatrix} \rho \\ \rho u \\ \rho E \end{bmatrix} \quad \text{and} \quad A(U) = \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{2}(\gamma-3)u^2 & (3-\gamma)u & \gamma-1 \\ \left(\frac{\gamma-1}{2}u^2 - H\right)u & H-(\gamma-1)u^2 & \gamma u \end{bmatrix}$$

The Jacombian matrix is diagonalizable: $A = P\Lambda P^{-1}$

$$\Lambda = \begin{pmatrix} u-a & 0 & 0 \\ 0 & u & 0 \\ 0 & 0 & u+a \end{pmatrix} \qquad P = \begin{pmatrix} 1 & 1 & 1 \\ u-a & u & u+a \\ H-au & u^2/2 & H+au \end{pmatrix} \qquad P^{-1} = \begin{pmatrix} \frac{1}{2}(\alpha_1 + \frac{u}{a}) & -\frac{1}{2}(\alpha_2 u + \frac{1}{a}) & \frac{\alpha_2}{2} \\ 1-\alpha_1 & \alpha_2 u & -\alpha_2 \\ \frac{1}{2}(\alpha_1 - \frac{u}{a}) & -\frac{1}{2}(\alpha_2 u - \frac{1}{a}) & \frac{\alpha_2}{2} \end{pmatrix}$$

with $\alpha_1 = (\gamma-1)u^2/(2a^2)$ and $\alpha_2 = (\gamma-1)/a^2$

Our PDE: $\dfrac{\partial r}{\partial t} + \dfrac{\partial r}{\partial x}\dfrac{dx}{dt} = 0$

$$\frac{\mathrm{d}r}{\mathrm{d}t} = \frac{\partial r}{\partial t} + \left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)\frac{\partial r}{\partial x}$$

(1) $\mathscr{C}$ ***is the characteristic curve where*** $\mathrm{d}x/\mathrm{d}t = \lambda$

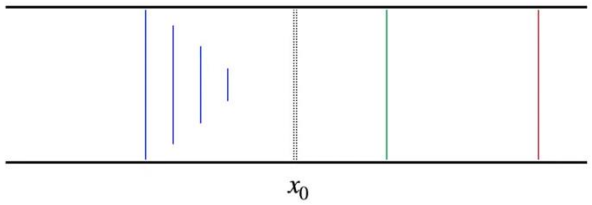
(2) $\left(\dfrac{\mathrm{d}r}{\mathrm{d}t}\right)_{\mathscr{C}} = 0$ ⟶ *r is constant over the characteristic curve* $\mathscr{C}$

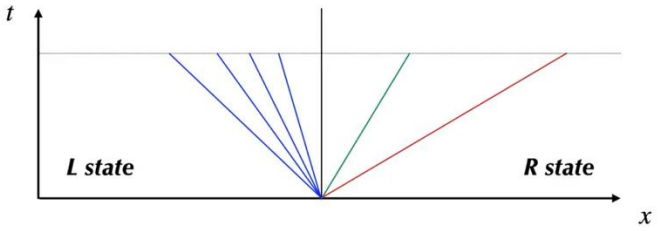
$$r^- = u - \frac{2a}{\gamma - 1} \quad \text{along} \quad \mathscr{C}^- : \frac{dx}{dt} = u - a$$

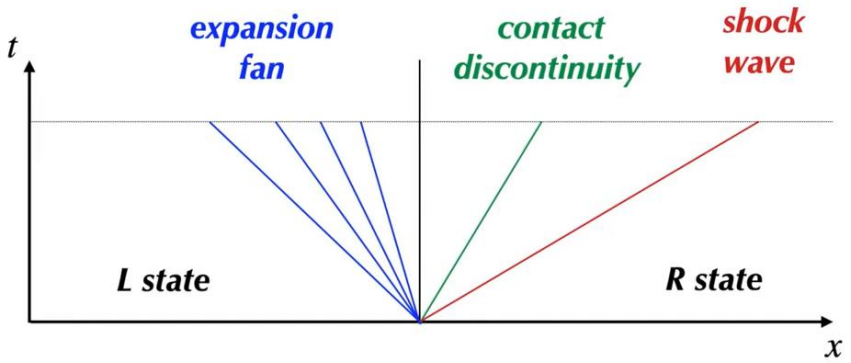
$$r^0 = p/\rho^\gamma \quad \text{along} \quad \mathscr{C}^0 : \frac{dx}{dt} = u$$

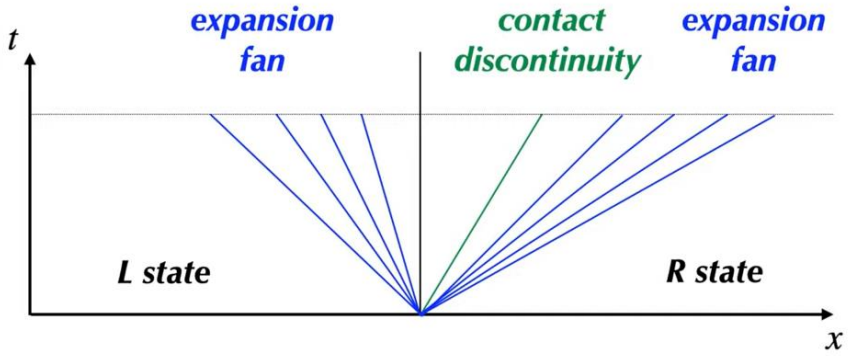
$$r^+ = u + \frac{2a}{\gamma - 1} \quad \text{along} \quad \mathscr{C}^+ : \frac{dx}{dt} = u + a$$

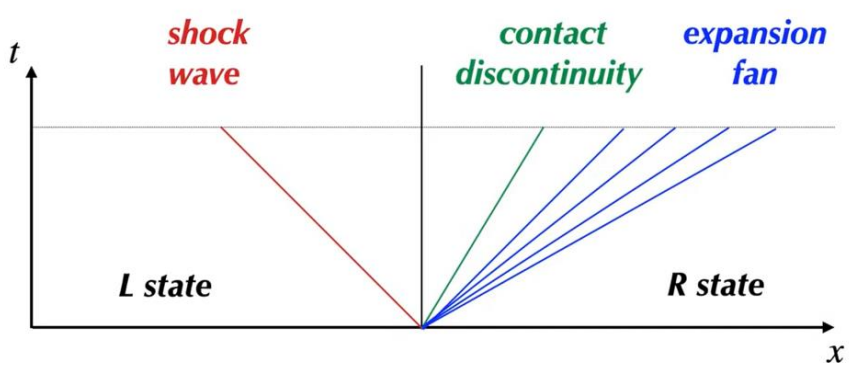
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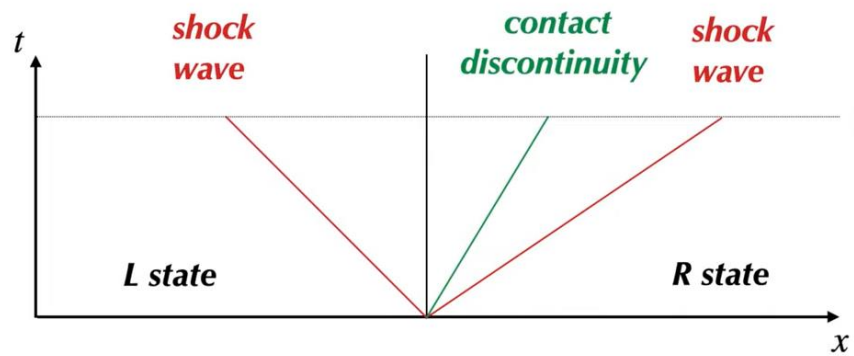




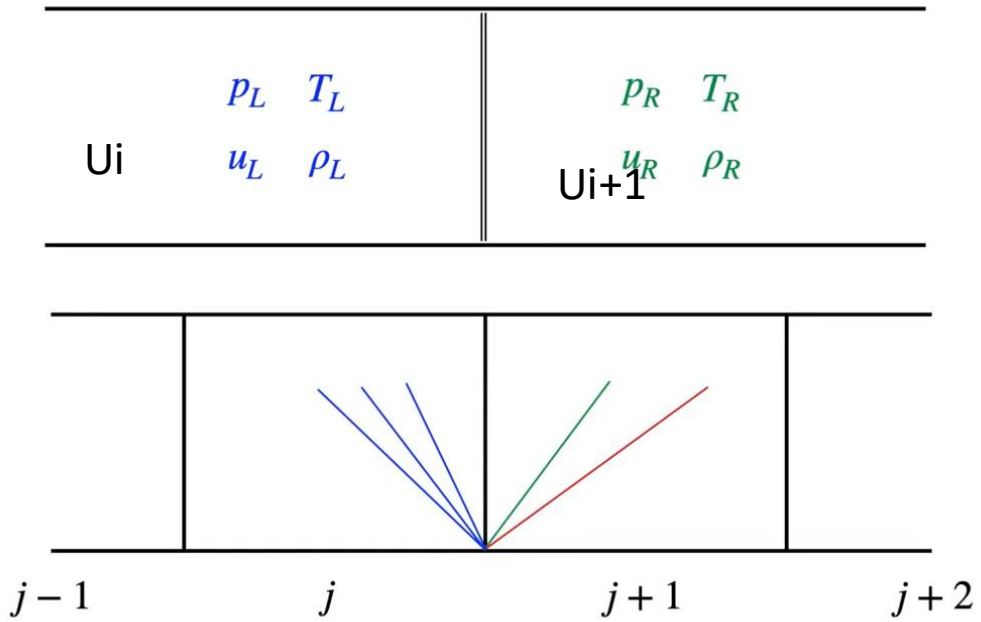




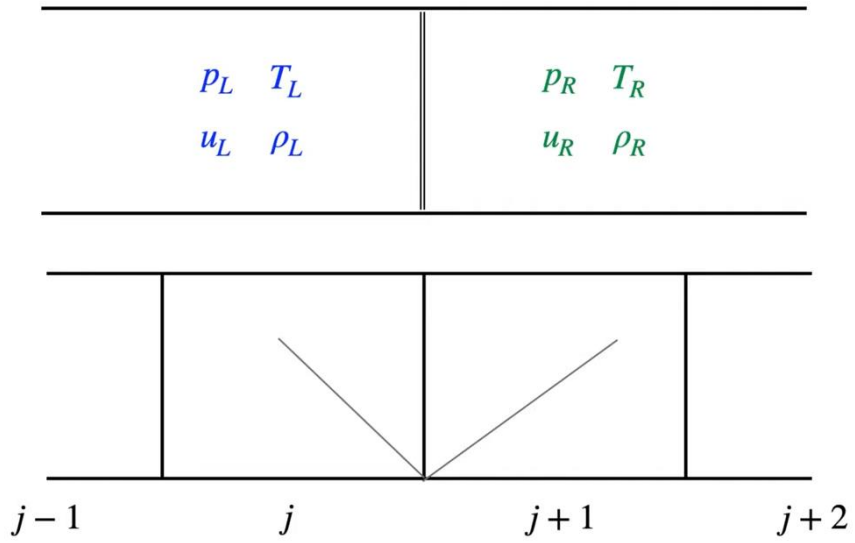




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Godunov, S. K. "A difference scheme for numerical solution of discontinuous solution of hydrodynamic equations." Math. Sbornik 47 (1959): 271-306.



Roe, P. L. "Approximate Riemann solvers, parameter vectors, and difference schemes." Journal of computational physics 43.2 (1981): 357-372.

TD - Computational Fluid Dynamics : Compressible flows (ROE scheme)

$$\frac{\partial \hat{U}}{\partial t} + \hat{A}_{j+1/2}\frac{\partial \hat{U}}{\partial x} = 0, \quad \hat{U}(x, n\Delta t) = \begin{cases} U_j^n, & x \leq (j+1/2)\Delta x \\ U_{j+1}^n, & x > (j+1/2)\Delta x \end{cases}$$

- ***The matrix is diagonalizable and has real eigenvalues*** $\Rightarrow$ $\hat{A}_{j+1/2} = \hat{P}_{j+1/2}\hat{\Lambda}_{j+1/2}\hat{P}^{-1}_{j+1/2}$
- ***The linear Riemann problem is consistent with the initial problem*** $\Rightarrow$ $\hat{A}_{j+1/2}(U, U) = A(U, U)$
- ***The numerical scheme is conservative*** $\Rightarrow$ $F(U) - F(V) = \hat{A}_{j+1/2}(U, V)(U - V)$

The Jacombian matrix is diagonalizable: $A = P\Lambda P^{-1}$

$$\Lambda = \begin{pmatrix} u-a & 0 & 0 \\ 0 & u & 0 \\ 0 & 0 & u+a \end{pmatrix} \qquad P = \begin{pmatrix} 1 & 1 & 1 \\ u-a & u & u+a \\ H-au & u^2/2 & H+au \end{pmatrix} \qquad P^{-1} = \begin{pmatrix} \frac{1}{2}(\alpha_1 + \frac{u}{a}) & -\frac{1}{2}(\alpha_2 u + \frac{1}{a}) & \frac{\alpha_2}{2} \\ 1-\alpha_1 & \alpha_2 u & -\alpha_2 \\ \frac{1}{2}(\alpha_1 - \frac{u}{a}) & -\frac{1}{2}(\alpha_2 u - \frac{1}{a}) & \frac{\alpha_2}{2} \end{pmatrix}$$

with $\alpha_1 = (\gamma-1)u^2/(2a^2)$ and $\alpha_2 = (\gamma-1)/a^2$

The new matrix is exactly equal to the Jacobian if we replace the variables (ρ, u, H) ***with the corresponding Roe's averages:***

$$\hat{\rho}_{j+1/2} = R_{j+1/2}\,\rho_j \qquad \hat{u}_{j+1/2} = \frac{R_{j+1/2}\,u_{j+1} + u_j}{1 + R_{j+1/2}} \qquad \hat{H}_{j+1/2} = \frac{R_{j+1/2}\,H_{j+1} + H_j}{1 + R_{j+1/2}} \qquad \hat{a}^2_{j+1/2} = (\gamma-1)\left(\hat{H}_{j+1/2} - \frac{u^2_{j+1/2}}{2}\right) \qquad R_{j+1/2} = \sqrt{\rho_{j+1}/\rho_j}$$

The new matrix is exactly equal to the Jacobian if we replace the variables (ρ, u, H) with the corresponding Roe's averages:

$$\hat{\rho}_{j+1/2} = R_{j+1/2}\,\rho_j \qquad \hat{u}_{j+1/2} = \frac{R_{j+1/2}\,u_{j+1} + u_j}{1 + R_{j+1/2}} \qquad \hat{H}_{j+1/2} = \frac{R_{j+1/2}\,H_{j+1} + H_j}{1 + R_{j+1/2}} \qquad \hat{a}^2_{j+1/2} = (\gamma - 1)\left(\hat{H}_{j+1/2} - \frac{u^2_{j+1/2}}{2}\right) \qquad R_{j+1/2} = \sqrt{\rho_{j+1}/\rho_j}$$

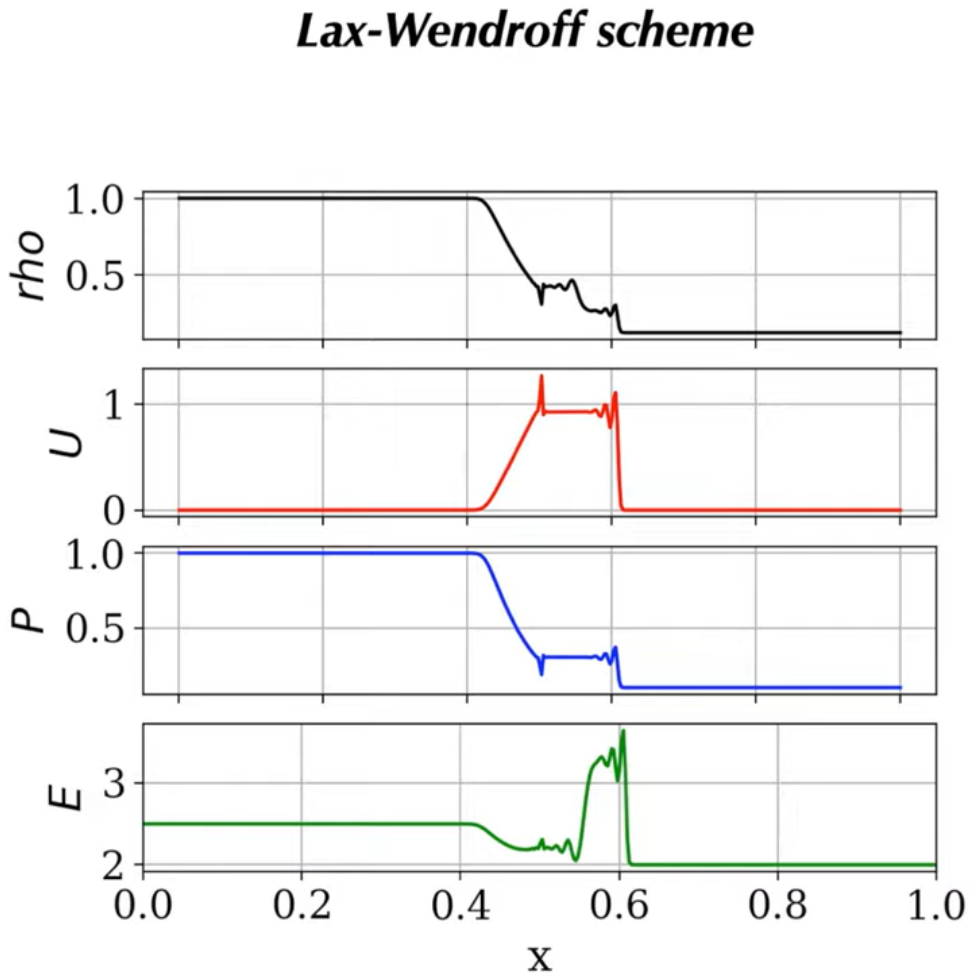
The Roe flux in the general conservative form becomes:

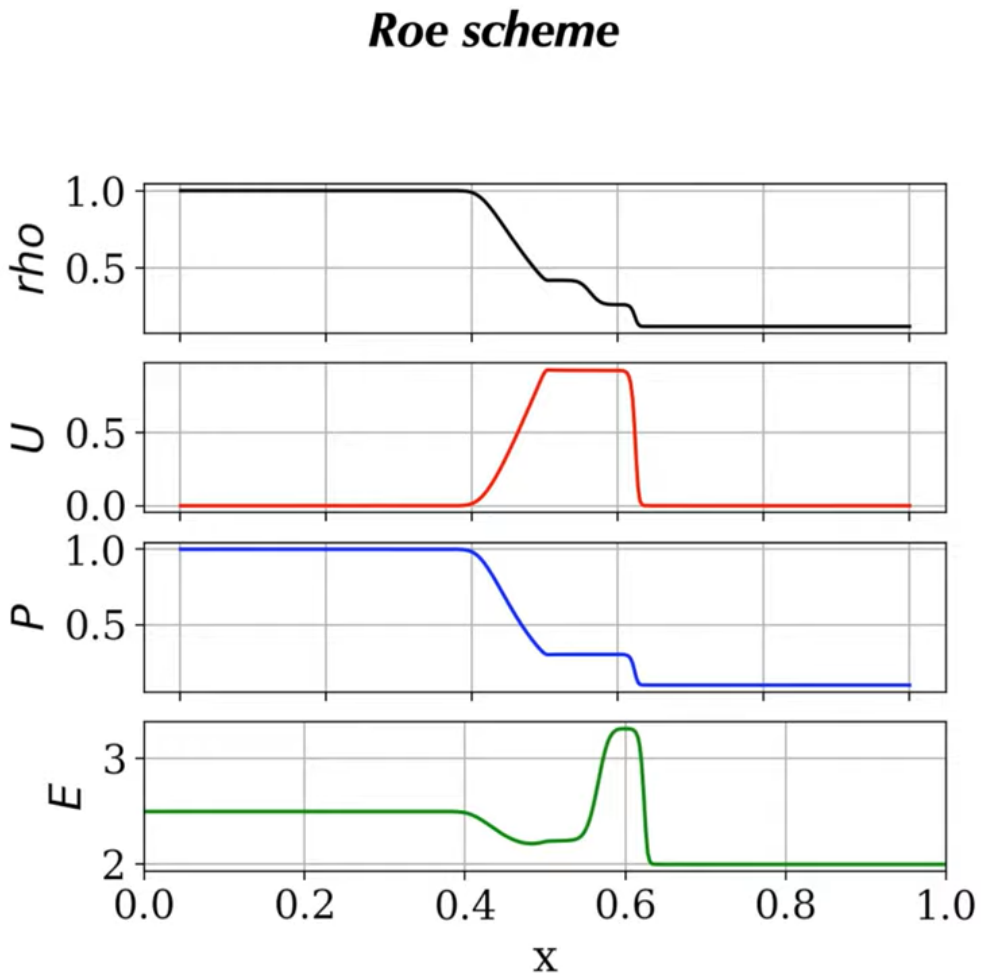
$$\left|\hat{A}_{j+1/2}\right| = \hat{P}_{j+1/2}\left|\hat{\Lambda}_{j+1/2}\right|\hat{P}^{-1}_{j+1/2}$$

$$\phi(U^n_j, U^n_{j+1}) = \frac{1}{2}[F(U^n_j) + F(U^n_{j+1})] - \frac{1}{2}\left|\hat{A}_{j+1/2}\right|(U^n_{j+1} - U^n_j)$$

$$U^{n+1}_j = U^n_j - \frac{\Delta t}{\Delta x}[\phi(U^n_j, U^n_{j+1}) - \phi(U^n_j, U^n_{j-1})]$$

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